

Chapter 7

- OpenMP allows for parallelization using multiple cores on a single node via threads with a shared memory model
- Relaxed Memory Model: Update memory at end, memory and cache not kept in sync, fix later
- Race Condition: Two processes change or read a variable at the same time, unsafe
- Private Variable (OpenMP): local and visible only to this thread, no race
- Shared Variable (OpenMP): non-local and visible to all threads, race possible
- Pragmas:
 - Parallel: defines a parallel region and spawns threads
 - For: Loop to share work
 - For reduction: Combines work at the end of a loop via reduction
 - Barrier: Makes all threads wait until all threads hit barrier
 - Masked / Master: Non-blocking and only runs on rank 0
 - Single: Runs on first thread to hit it, blocks at end
 - Critical / Atomic: Used to avoid race conditions
- Get number of threads via `omp_get_num_threads()`
- Set number of threads via `omp_set_num_threads(int threads)`
- Thread Id via `omp_get_thread_num()`, but must be in parallel region!
- Loop level OpenMP is useful when a quick solution is needed and minimal setup is needed
 - Avoid this if the parallel region will be rapidly setup and torn down!
- High level OpenMP is useful when a whole program is parallelized
 - Startup costs associated with spawning threads is avoided
 - Programming difficulty is increased to avoid race conditions between non-performant parts of the code
- MPI can be mixed with OpenMP if needed

Core binding can be controlled if needed and the default is bad

- Ex:
`export OMP_PLACES=cores`
`export OMP_CPU_BIND=true`

Memory will be allocated by "first-touch" so it can be helpful not to allocate and assign memory until the threads are created which will access that data, such that it ends up local.

- May matter with NUMA nodes for example where not all memory can be accessed at the same speed

Since race conditions can exist, code must be checked for correctness after each parallel region

- This is done by comparing with a version of the code which is not parallel
- Small numerical issues may not be problematic, helpful to define a tolerance if lots of floating point math is occurring for the test